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TEACHER TRAINING PROGRAM

3D PRINTING FOR EDUCATORS

FROM IDEA TO CLASSROOM

Empowering teachers to design and create educational resources using 3D printing — a practical, hands-on program built for educators with no prior engineering or CAD experience.

FROM IDEA TO CLASSROOM

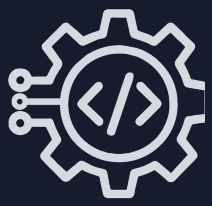
This training program is designed with a practical, hands-on approach to equip teachers with the essential skills needed to integrate 3D printing into their classrooms.

The program focuses on enabling educators to independently design, prepare, and produce educational resources using simple, accessible tools that require no prior engineering or CAD experience. By the end of the training, participants will have the confidence to create customized teaching aids and classroom models that enhance student engagement and support interactive learning.

5
TRAINING MODULES

2
CORE PLATFORMS USED

0
PRIOR CAD EXPERIENCE REQUIRED



Program Objectives

Upon successful completion of the program, participants will be able to understand 3D printing fundamentals, safely operate professional equipment, design original models, and produce classroom-ready educational resources — detailed on the next page.

WHAT PARTICIPANTS WILL BE ABLE TO DO

Upon successful completion of the program, participants will be able to:

- Understand the fundamentals of 3D printing technology.
- Safely operate the Bambu Lab A1 3D printer.
- Prepare models for printing using Bambu Studio.
- Design simple educational models using Tinkercad.
- Export designs in STL format.
- Configure print settings according to project requirements.
- Produce high-quality prints and troubleshoot common printing issues.

WHY THESE TOOLS

WHY TINKERCAD?

Tinkercad has been selected because it is ideally suited for educators who are new to 3D design. Rather than spending time learning complex engineering software, teachers can focus on transforming their ideas into practical educational resources.

- A completely free platform.
- No software installation required.
- Browser-based access from any computer.
- An intuitive interface designed for beginners.
- A short learning curve, allowing teachers to become productive within a few hours.
- All the essential tools required to create classroom-ready educational models.

WHY BAMBU LAB A1?

The Bambu Lab A1 has been selected as the training platform because it provides an excellent balance of simplicity, reliability, and performance for educational environments.

- Easy operation.
- Fast and high-quality printing.
- Minimal maintenance requirements.
- Automatic calibration features that reduce user errors.
- An ideal experience for first-time users and educational institutions.

FIVE MODULES, ONE PATH

The program follows a highly practical, project-based learning approach that emphasizes hands-on experience over theory, structured across five progressive modules.

3D DESIGN WITH TINKERCAD

01

Participants will learn how to:

- Navigate the Tinkercad workspace.
- Create basic 3D objects.
- Apply accurate dimensions.
- Align and organize objects.
- Combine and subtract shapes.
- Add 3D text.
- Design simple educational models for classroom use.

PREPARING MODELS FOR PRINTING

02

Topics include:

- Exporting STL files.
- Using Bambu Studio.
- Selecting appropriate print settings.
- Sending models to the printer.
- Monitoring the printing process.

MODULES 3 & 4

INTRODUCTION TO 3D PRINTING

03

Participants will explore:

- What is 3D printing?
- Educational applications of 3D printing.
- Types of printing materials (filaments).
- Components of a 3D printer.
- Safety procedures and best practices.

OPERATING THE BAMBU LAB A1

04

Hands-on training covering:

- Printer overview.
- Loading filament.
- Automatic calibration.
- Starting a print.
- Removing finished models safely.
- Routine maintenance and care.

MODULE 5 & METHODOLOGY

PRACTICAL CLASSROOM PROJECT

05

Each participant will design and print an educational model related to their teaching subject. Possible examples include:

- Geometric models.
- Science teaching aids.
- Classroom nameplates.
- Tool holders and organizers.
- Educational demonstrations.
- Subject-specific learning resources.

05 TRAINING METHODOLOGY

The program follows a highly practical, project-based learning approach that emphasizes hands-on experience over theory. Training includes:

- Concise theoretical introductions.
- Live demonstrations.
- Guided practical exercises.
- Individual hands-on practice with the equipment.
- Real classroom-oriented design and printing projects.

BY THE END OF THE PROGRAM

By the end of the program, every participant will be able to:

- Design a simple 3D model independently.
- Prepare it for printing using Bambu Studio.
- Print it successfully on the Bambu Lab A1.
- Apply these skills to create customized educational resources for their own classrooms.

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Bring precision-built, project-based 3D printing training to your school or institute.

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